

# Practical 1

**Aim:**Linux System Administration and File Management for Marketing Analytics Team

**Problem definition:**

Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Tasks:

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.
10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

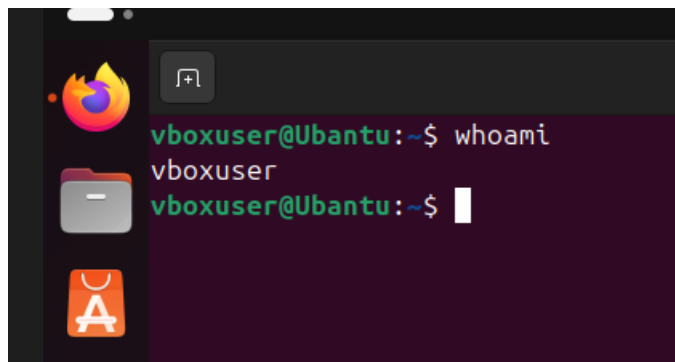
### Objectives:

- To learn basic Linux system administration commands.
- To create and manage directories and files for a project environment.
- To perform file operations such as creating, copying, moving, renaming, and deleting files.
- To search for files and specific content using Linux commands.
- To monitor system information, uptime, and resource usage.
- To understand the importance of file permissions and data security.
- To create backups of important project files.
- To gain practical experience with Bash shell commands used in enterprise environments.

### Observation / Output:

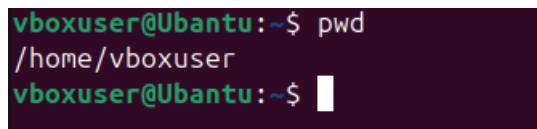
#### Tasks:

1. Identify the current user.

A terminal window with a dark purple background. On the left sidebar, there are icons for Firefox, a file manager, and the Ubuntu Software Center. The terminal text shows the prompt 'vboxuser@Ubuntu:~\$' followed by the command 'whoami'. The output is 'vboxuser' on the next line, followed by another prompt 'vboxuser@Ubuntu:~\$' with a cursor.

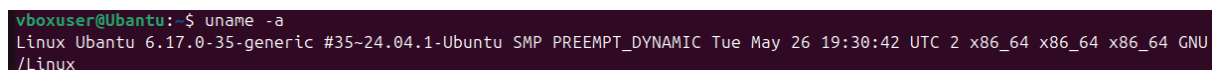
```
vboxuser@Ubuntu:~$ whoami
vboxuser
vboxuser@Ubuntu:~$
```

2. Find the current working directory.

A terminal window with a dark purple background. The terminal text shows the prompt 'vboxuser@Ubuntu:~\$' followed by the command 'pwd'. The output is '/home/vboxuser' on the next line, followed by another prompt 'vboxuser@Ubuntu:~\$' with a cursor.

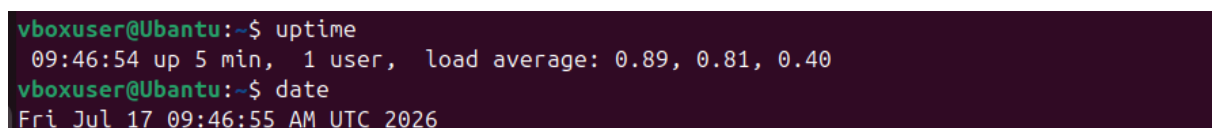
```
vboxuser@Ubuntu:~$ pwd
/home/vboxuser
vboxuser@Ubuntu:~$
```

3. Display system information.

A terminal window with a dark purple background. The terminal text shows the prompt 'vboxuser@Ubuntu:~\$' followed by the command 'uname -a'. The output is a single line of system information: 'Linux Ubuntu 6.17.0-35-generic #35-24.04.1-Ubuntu SMP PREEMPT\_DYNAMIC Tue May 26 19:30:42 UTC 2 x86\_64 x86\_64 x86\_64 GNU/Linux'.

```
vboxuser@Ubuntu:~$ uname -a
Linux Ubuntu 6.17.0-35-generic #35-24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue May 26 19:30:42 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

4. Check system uptime and date.

A terminal window with a dark purple background. The terminal text shows the prompt 'vboxuser@Ubuntu:~\$' followed by the command 'uptime'. The output is '09:46:54 up 5 min, 1 user, load average: 0.89, 0.81, 0.40'. Below that, the prompt 'vboxuser@Ubuntu:~\$' is followed by the command 'date', with the output 'Fri Jul 17 09:46:55 AM UTC 2026'.

```
vboxuser@Ubuntu:~$ uptime
09:46:54 up 5 min, 1 user, load average: 0.89, 0.81, 0.40
vboxuser@Ubuntu:~$ date
Fri Jul 17 09:46:55 AM UTC 2026
```

5. Create a directory named MarketingAnalytics.

```
vboxuser@Ubuntu:~$ mkdir MarketingAnalytics
vboxuser@Ubuntu:~$ cd MarketingAnalytics
vboxuser@Ubuntu:~/MarketingAnalytics$
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ mkdir RawData ProcessedData Reports
vboxuser@Ubuntu:~/MarketingAnalytics$ ls
ProcessedData RawData Reports
vboxuser@Ubuntu:~/MarketingAnalytics$
```

7. Create data and report files.

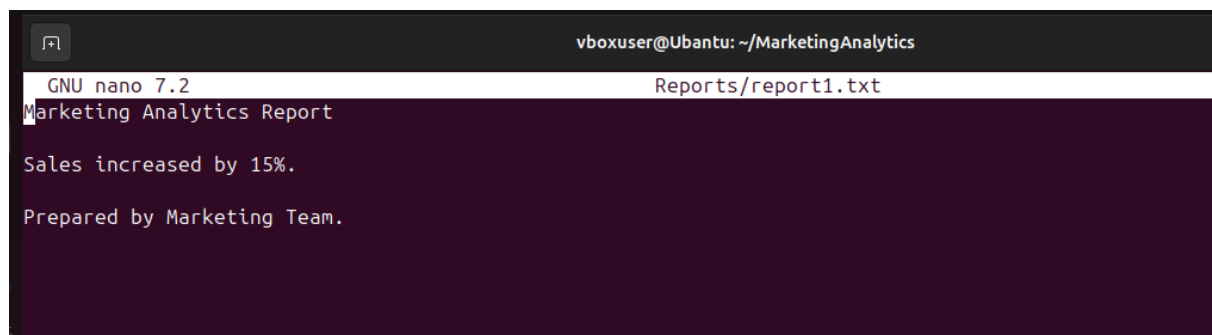
```
vboxuser@Ubuntu:~/MarketingAnalytics$ touch RawData/customer_data.txt\
> touch Reports/report1.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

8. Add sample content.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ nano RawData/customer_data.txt
vboxuser@Ubuntu:~/MarketingAnalytics$ nano Reports/report1.txt
```



```
vboxuser@Ubuntu: ~/MarketingAnalytics
GNU nano 7.2 RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
```



```
vboxuser@Ubuntu: ~/MarketingAnalytics
GNU nano 7.2 Reports/report1.txt
Marketing Analytics Report

Sales increased by 15%.

Prepared by Marketing Team.
```

9. Display file contents.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ cat RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
vboxuser@Ubuntu:~/MarketingAnalytics$ cat Reports/report1.txt
Marketing Analytics Report

Sales increased by 15%.

Prepared by Marketing Team.
```

10. List all files and directories.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ ls -R
.:
ProcessedData RawData Reports

./ProcessedData:

./RawData:
customer_data.txt customer_data.txttouch

./Reports:
report1.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

11. Search for report files.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ find . -name "*.txt"
./RawData/customer_data.txt
./Reports/report1.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

12. Search specific content inside files.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ grep "Sales" Reports/report1.txt
Sales increased by 15%.
```

13. Copy report files to Backup directory.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ mkdir Backup
vboxuser@Ubuntu:~/MarketingAnalytics$ cp Reports/report1.txt Backup/
vboxuser@Ubuntu:~/MarketingAnalytics$ ls Backup
report1.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

14. Rename files.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ mv Reports/report1.txt Reports/final_report.txt
vboxuser@Ubuntu:~/MarketingAnalytics$ ls Reports
final_report.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

15. Move files between directories.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ mv RawData/customer_data.txt ProcessedData/
vboxuser@Ubuntu:~/MarketingAnalytics$ ls ProcessedData
customer_data.txt
vboxuser@Ubuntu:~/MarketingAnalytics$
```

16. Delete a file.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ rm Backup/report1.txt
vboxuser@Ubuntu:~/MarketingAnalytics$ ls Backup
vboxuser@Ubuntu:~/MarketingAnalytics$
```

17. Delete an empty directory.

```
vboxuser@Ubuntu:~/MarketingAnalytics$ rmdir RawData
vboxuser@Ubuntu:~/MarketingAnalytics$ ls RawData
ls: cannot access 'RawData': No such file or directory
vboxuser@Ubuntu:~/MarketingAnalytics$
```

## Result :

The Linux project environment for the **Marketing Analytics** team was successfully created. Directories and files were organized efficiently, sample data was added, and file operations such as copy, move, rename, and delete were performed successfully. System information, uptime, and resource usage were verified. Backup directories were created, and report files were successfully copied, demonstrating effective Linux system administration and file management.

## Key Questions /Analysis:

Q1. What is the difference between absolute and relative paths?

Answer:

- **Absolute Path:** Specifies the complete path from the root (/) directory.
  - Example: /home/student/MarketingAnalytics/Reports/report1.txt

- **Relative Path:** Specifies the location relative to the current working directory.
  - Example: Reports/report1.txt

**Difference:**

| <b>Absolute Path</b>           | <b>Relative Path</b>          |
|--------------------------------|-------------------------------|
| Starts from / (root directory) | Starts from current directory |
| Works from anywhere            | Depends on current location   |
| Longer                         | Shorter and convenient        |

Q2. Why are Linux permissions important for file security?

**Answer:**

Linux permissions protect files and directories from unauthorized access. They control who can **read (r)**, **write (w)**, and **execute (x)** files.

Benefits:

- Prevent unauthorized modifications.
- Protect sensitive company data.
- Improve system security.
- Allow controlled access for different users and groups.

Q3. What is the purpose of the grep command?

**Answer:**

The grep command searches for specific text or patterns inside files.

Example:

```
grep "Sales" report1.txt
```

It helps users quickly locate required information from large files or multiple documents.

Q4. How does Linux support efficient file management?

**Answer:**

Linux provides powerful commands for managing files efficiently.

Features include:

- Creating directories (mkdir)
- Copying files (cp)
- Moving files (mv)

- Renaming files (mv)
- Deleting files (rm)
- Searching files (find)
- Searching content (grep)
- Viewing permissions (ls -l)
- Compressing backups (tar, gzip)

These utilities make file organization and administration simple and efficient.

Q5. Why are backups important in enterprise environments?

**Answer:**

Backups are essential because they:

- Prevent permanent data loss.
- Recover files after accidental deletion.
- Protect against hardware failures.
- Support disaster recovery.
- Ensure business continuity.
- Help restore important project data quickly.

Regular backups reduce downtime and improve data reliability.

### Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
vboxuser@Ubuntu:~/BDA/Practical1$ nano create_project.sh
vboxuser@Ubuntu:~/BDA/Practical1$ chmod +x create_project.sh
vboxuser@Ubuntu:~/BDA/Practical1$ ./create_project.sh
Marketing Analytics project created successfully.
vboxuser@Ubuntu:~/BDA/Practical1$
```

2. Find the top 10 largest files in the system.

```
vboxuser@Ubuntu:~/BDA/Practical1$ sudo find / -type f -exec du -h {} + 2>/dev/nul
ll | sort -rh | head -10
[sudo] password for vboxuser:
697M    /home/vboxuser/hadoop-3.3.6.tar.gz
506M    /var/lib/snapd/seed/snaps/gnome-42-2204_176.snap
297M    /home/vboxuser/hadoop/share/hadoop/tools/lib/aws-java-sdk-bundle-1.12.36
7.jar
270M    /var/lib/snapd/seed/snaps/firefox_4173.snap
139M    /snap/firefox/4173/usr/lib/firefox/libxul.so
137M    /usr/lib/x86_64-linux-gnu/libLLVM.so.20.1
137M    /usr/lib/jvm/java-11-openjdk-amd64/lib/modules
123M    /usr/lib/jvm/java-17-openjdk-amd64/lib/modules
118M    /usr/lib/llvm-18/lib/libLLVM.so.1
112M    /snap/gnome-42-2204/176/usr/lib/x86_64-linux-gnu/libLLVM-15.so.1
```

3. Schedule a backup task using cron.

```
vboxuser@Ubuntu:~/BDA/Practical1$ crontab -e
no crontab for vboxuser - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
vboxuser@Ubuntu:~/BDA/Practical1$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * cp -r ~/MarketingAnalytics ~/Backup
```

4. Monitor CPU utilization continuously using top/htop.

**Top:**

```
top - 15:46:05 up 6 min, 1 user, load average: 0.87, 1.01, 0.53
Tasks: 210 total, 1 running, 208 sleeping, 0 stopped, 1 zombie
%Cpu(s):  9.2 us, 17.6 sy,  0.0 ni, 69.7 id,  0.4 wa,  0.0 hi,  3.1 si,  0.0 st
MiB Mem : 3914.8 total, 149.6 free, 1977.3 used, 2126.2 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 1937.4 avail Mem
```

| PID  | USER     | PR | NI  | VIRT    | RES    | SHR    | S | %CPU | %MEM | TIME+   | COMMAND                         |
|------|----------|----|-----|---------|--------|--------|---|------|------|---------|---------------------------------|
| 1867 | vboxuser | 20 | 0   | 4037376 | 421240 | 161656 | S | 28.1 | 10.5 | 1:07.51 | gnome-shell                     |
| 2525 | vboxuser | 20 | 0   | 10.9g   | 347344 | 160092 | S | 17.2 | 8.7  | 1:01.17 | firefox                         |
| 3097 | vboxuser | 20 | 0   | 2893540 | 482864 | 106672 | S | 10.9 | 12.0 | 0:36.03 | Isolated Web Co                 |
| 674  | root     | 20 | 0   | 1849232 | 44176  | 24816  | S | 5.6  | 1.1  | 0:04.62 | snappd                          |
| 37   | root     | 20 | 0   | 0       | 0      | 0      | I | 3.3  | 0.0  | 0:01.24 | kworker/u9:1-events_unbound     |
| 3361 | vboxuser | 20 | 0   | 556516  | 57016  | 44792  | S | 3.3  | 1.4  | 0:02.51 | gnome-terminal-                 |
| 24   | root     | 20 | 0   | 0       | 0      | 0      | S | 0.3  | 0.0  | 0:01.30 | ksoftirqd/1                     |
| 193  | root     | 20 | 0   | 0       | 0      | 0      | S | 0.3  | 0.0  | 0:00.41 | jbd2/sda2-8                     |
| 1969 | vboxuser | 20 | 0   | 388648  | 12512  | 7232   | S | 0.3  | 0.3  | 0:00.87 | ibus-daemon                     |
| 2712 | vboxuser | 20 | 0   | 2459384 | 127732 | 92792  | S | 0.3  | 3.2  | 0:02.31 | Privileged Cont                 |
| 1    | root     | 20 | 0   | 23324   | 14672  | 9848   | S | 0.0  | 0.4  | 0:02.41 | systemd                         |
| 2    | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.06 | kthread                         |
| 3    | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.00 | pool_workqueue_release          |
| 4    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-rcu_gp                |
| 5    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-sync_wq               |
| 6    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-kvfree_rcu_reclaim    |
| 7    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-slub_flushwq          |
| 8    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-netns                 |
| 9    | root     | 20 | 0   | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.85 | kworker/0:0-events              |
| 11   | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.57 | kworker/0:0H-kblockd            |
| 12   | root     | 20 | 0   | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/u8:0                    |
| 13   | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker/R-mm_percpu_wq          |
| 14   | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.49 | ksoftirqd/0                     |
| 15   | root     | 20 | 0   | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.78 | rcu_preempt                     |
| 16   | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.00 | rcu_exp_par_gp_kthread_worker/0 |

5. Generate a disk usage report for all project directories.

```
vboxuser@Ubuntu:~/BDA/Practical1$ du -sh ~/MarketingAnalytics/*
4.0K   /home/vboxuser/MarketingAnalytics/Backup
8.0K   /home/vboxuser/MarketingAnalytics/ProcessedData
8.0K   /home/vboxuser/MarketingAnalytics/Reports
vboxuser@Ubuntu:~/BDA/Practical1$ du -sh ~/MarketingAnalytics/* > ~/disk_usage_report.txt
vboxuser@Ubuntu:~/BDA/Practical1$ cat ~/disk_usage_report.txt
4.0K   /home/vboxuser/MarketingAnalytics/Backup
8.0K   /home/vboxuser/MarketingAnalytics/ProcessedData
8.0K   /home/vboxuser/MarketingAnalytics/Reports
vboxuser@Ubuntu:~/BDA/Practical1$
```

## Tools /Technology used:

Ubuntu Linux / CentOS Linux

Linux Terminal

Bash Shell

## Conclusion:

This practical provided hands-on experience with Linux system administration and file management. Various Linux commands were used to manage users, directories, files,

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permissions, and backups. System information and resource usage were monitored, while search and file management utilities improved efficiency. The supplementary tasks demonstrated automation using shell scripts, scheduled backups with cron, CPU monitoring, and disk usage analysis. Overall, the practical strengthened the understanding of Linux administration skills required in enterprise environments such as DataAnalytics Inc.